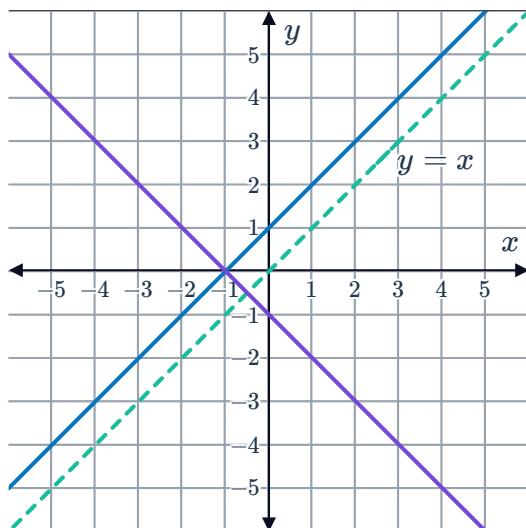


Inverse functions

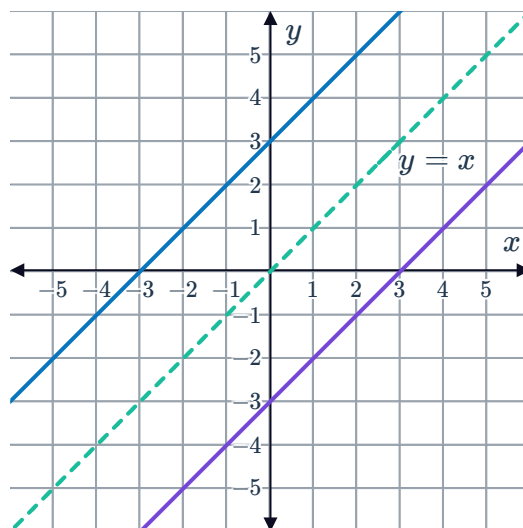


- 1 Explain how a graphing utility such as a graphing calculator be used to visually determine if two functions are inverses of each other.
- 2 In each graph below two functions and the line $y = x$ are drawn. For each graph, state whether the functions are inverse functions of each other.

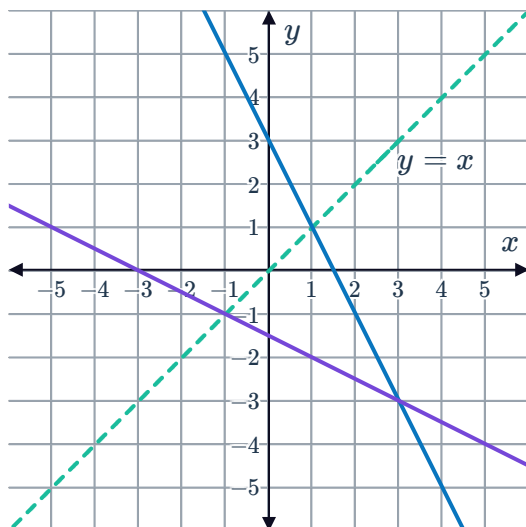
a



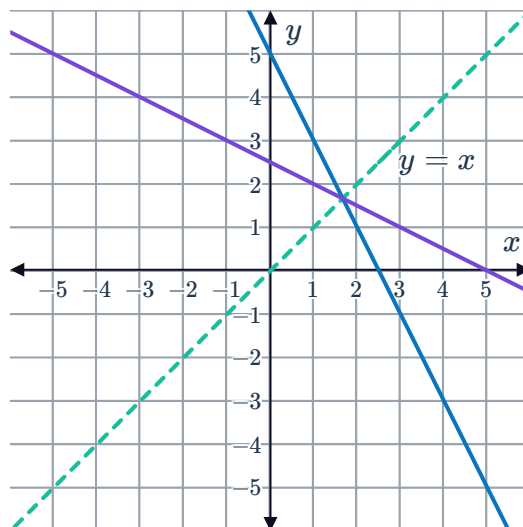
b



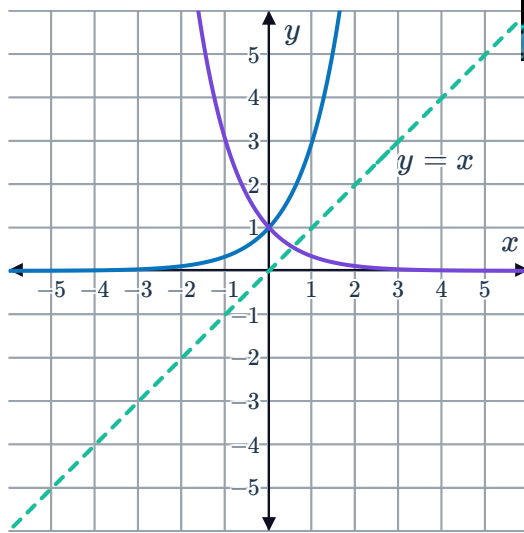
c



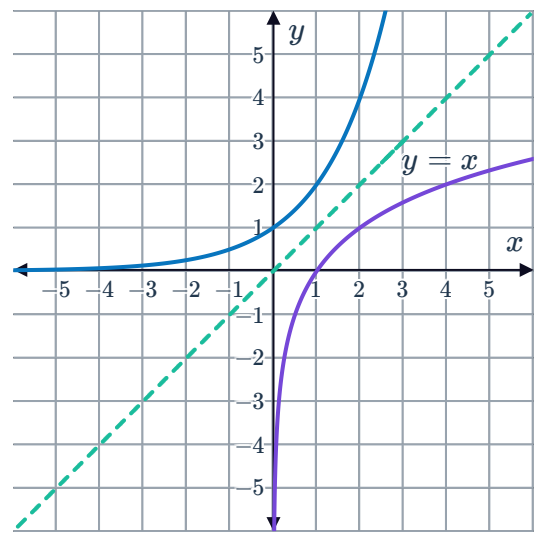
d



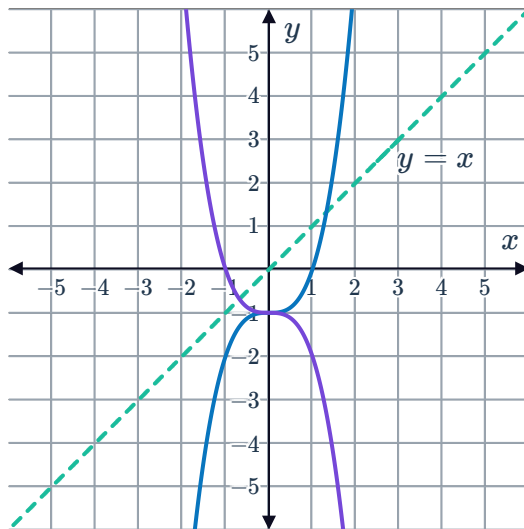
e



f



g



3 For each of the following functions, find an expression for the inverse function $y = f^{-1}(x)$:

a $f(x) = 7x - 8$

b $f(x) = \frac{4x}{3} - 5$

c $f(x) = 6x^3$

d $f(x) = \sqrt[3]{x-2}$

e $f(x) = \frac{8}{x-9} + 5$

f $f(x) = (x-4)^3 + 3$

g $f(x) = \frac{8}{5x+3}$

h $f(x) = \frac{7}{(x+8)^3} - 3$

i $f(x) = 4(9x-2)^3 - 7$

4 For each of the following functions, find:

i $f^{-1}(x)$

ii The domain of $f^{-1}(x)$

iii The range of $f^{-1}(x)$

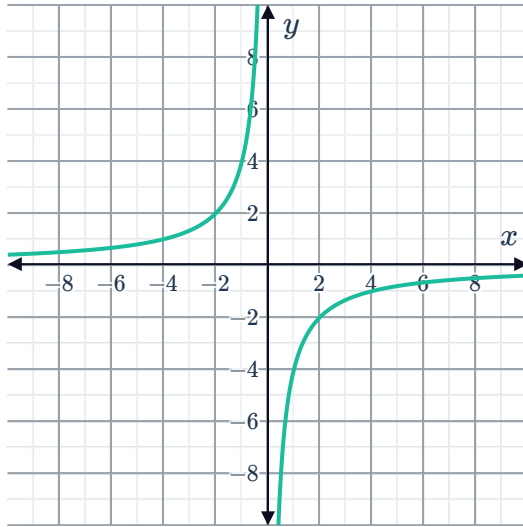
a $f(x) = 8x - 9$ defined over $[-4, 2]$

b $f(x) = x^2$ defined over $[0, \infty)$

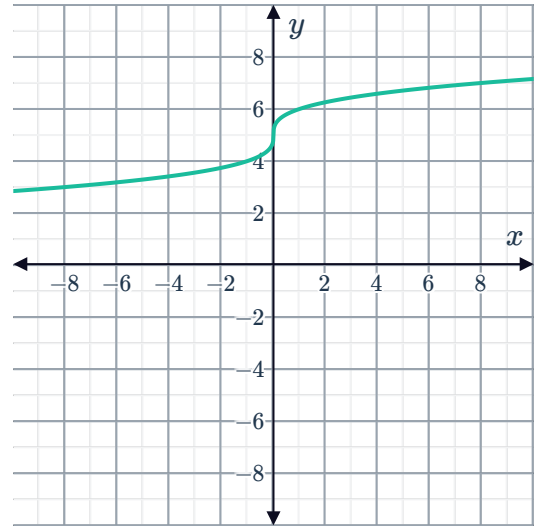


- 5 For each of the following functions, find an expression for the inverse function $y = f^{-1}(x)$:

a $f(x) = -\frac{4}{x}$



b $f(x) = \sqrt[3]{x} + 5$

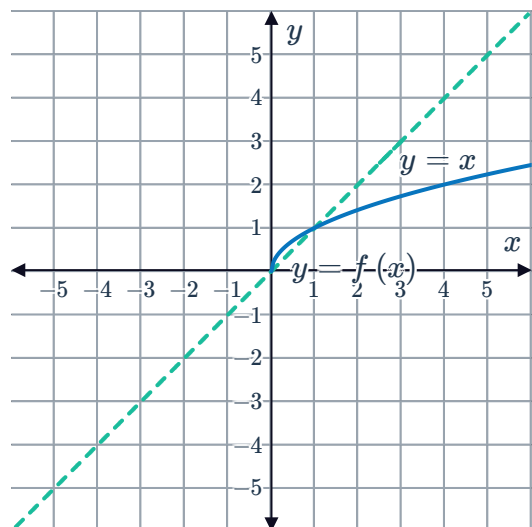


Graphs of inverse functions

- 6 Consider the functions $f(x) = \frac{1}{x} + 3$ and $g(x) = \frac{1}{x-3}$.

- a Sketch the graph of $f(x)$.
- b Sketch the graph of $g(x)$ on the same set of axes.
- c Are $f(x)$ and $g(x)$ inverses?

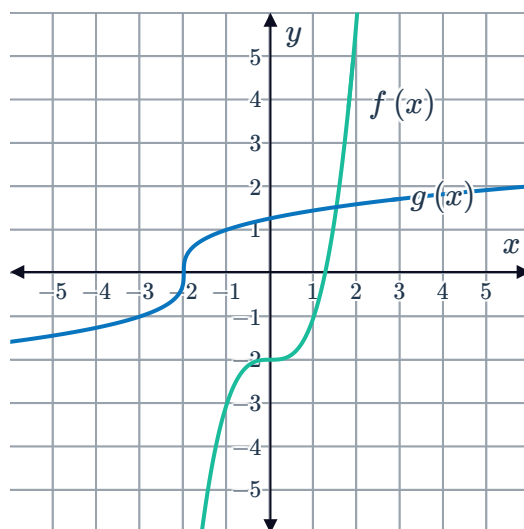
- 7 Consider the graph of the function $f(x)$ over the line $y = x$:
Sketch the graph of $f^{-1}(x)$.



8 Consider the graphs of $f(x)$ and $g(x)$:

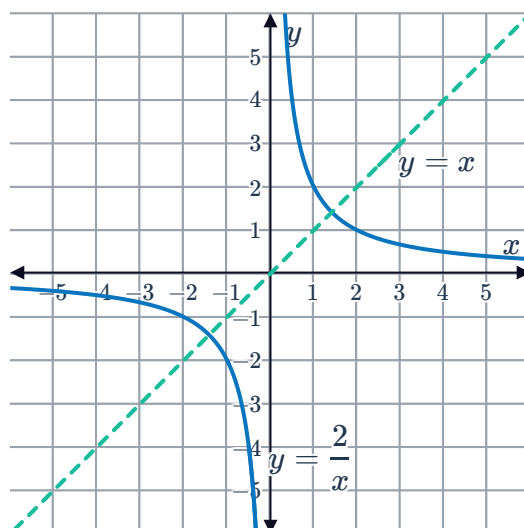


- State the equation of $f(x)$.
- State the equation of $g(x)$.
- Evaluate $f(g(x))$.
- Evaluate $g(f(x))$.
- State whether the following statements are correct:
 - $g(x)$ is an inverse of $f(x)$.
 - $f(g(x))$ has gradient -2 .
 - $f(x)$ is an inverse of $g(x)$.
 - $g(f(x))$ has gradient 1 .



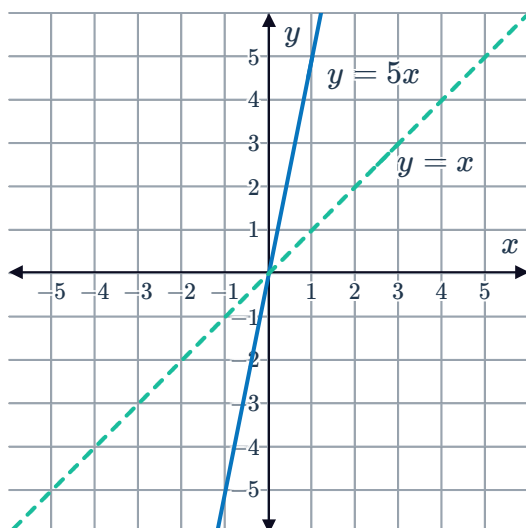
9 Consider the graph of $y = \frac{2}{x}$ over the line $y = x$:

- Sketch the graph of the inverse of $y = \frac{2}{x}$.
- Compare the inverse graph to the original graph.

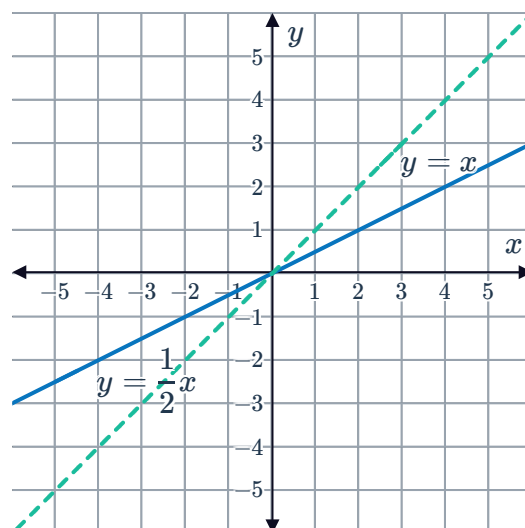


10 Sketch the graph of the inverse of the following functions:

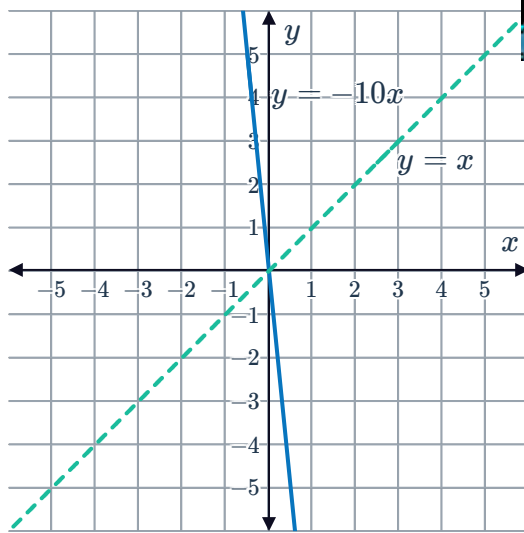
a



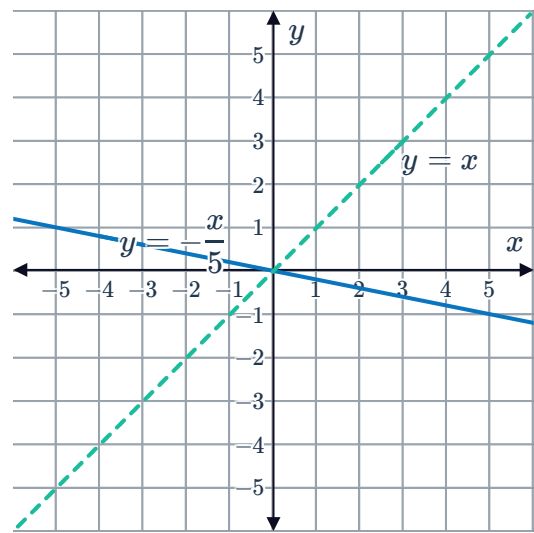
b



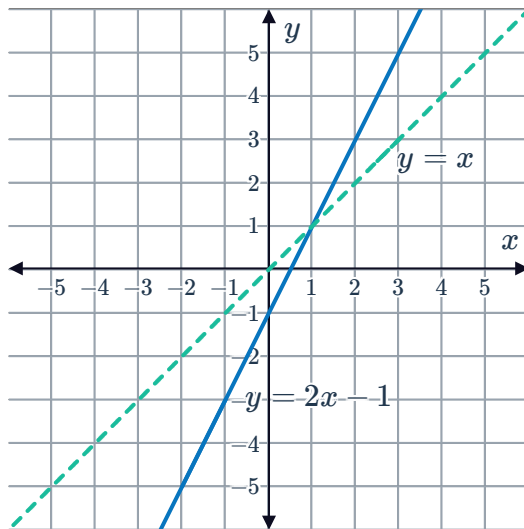
c



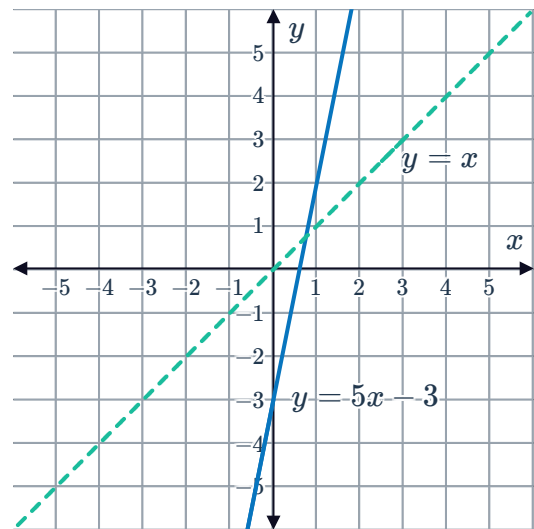
d



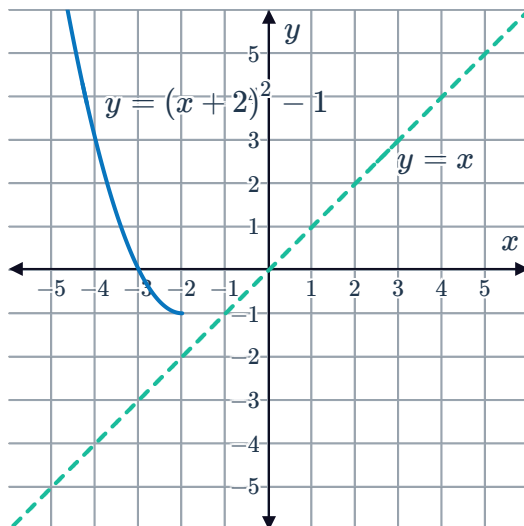
e



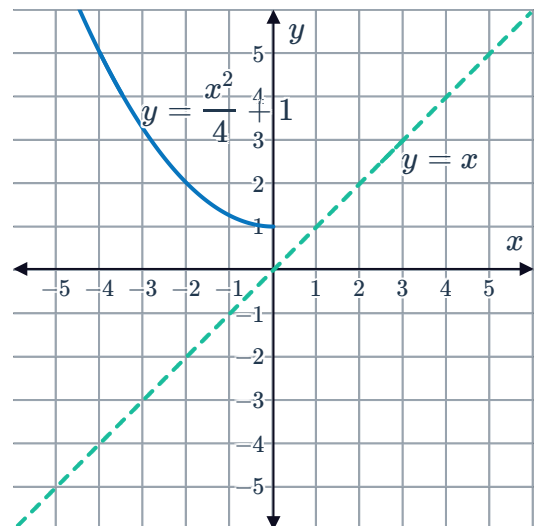
f



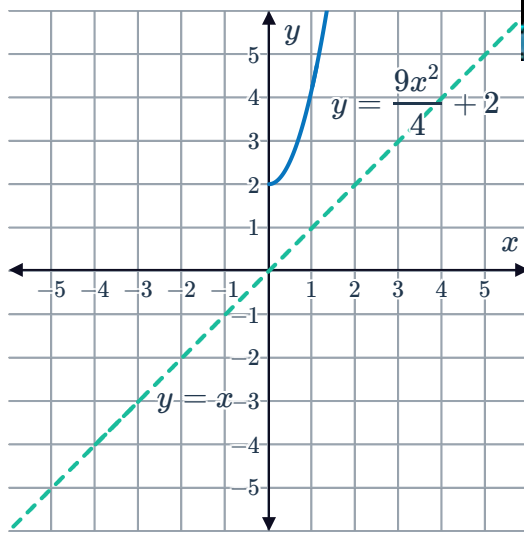
g



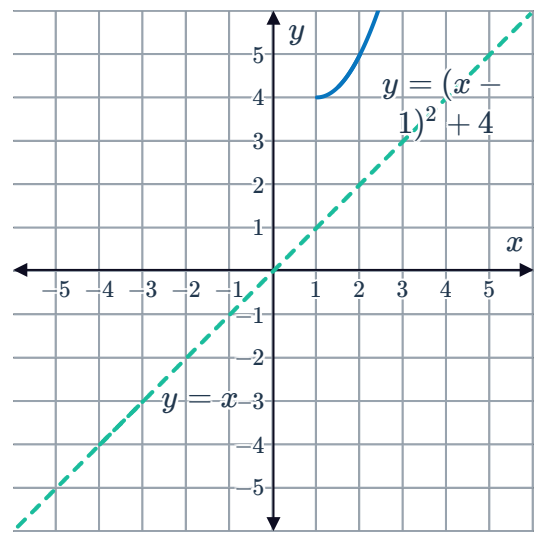
h



i



j



11 For each of the following functions:

- i Sketch the function $f(x)$ over its domain.
- ii Find the inverse, f^{-1} .
- iii State the domain of f^{-1} .
- iii State the range of f^{-1} .
- iv Sketch the function f^{-1} over its domain.

- a $f(x) = x + 6$ defined over the interval $[0, \infty)$.
- b $f(x) = 7 - x$ defined over the interval $[2, 9]$.
- c $f(x) = (x - 6)^2 - 2$ defined over the interval $[6, \infty)$.
- d $f(x) = \sqrt{4 - x}$ defined over the interval $[0, 4]$.
- e $f(x) = (x + 2)^2 + 3$ defined over the interval $[0, \infty)$.

12 Consider the functions $f(x) = x^2 - 5$ and $g(x) = \sqrt{x + 5}$, for $x \geq 0$. The function y is defined as $y = g(f(x))$, for $x \geq 0$.

- a State the equation for y .
- b Graph the functions $f(x)$, $g(x)$ and y on the same set of axes.
- c What do you notice about the graph of y in relation to the graphs of $f(x)$ and $g(x)$?



Applications

- 13** The function $t = \sqrt{\frac{d}{4.9}}$ can be used to find the number of seconds it takes for an object in Earth's atmosphere to fall d metres.

- a** State the function for d in terms of t .
- b** Find the distance a skydiver has fallen 5 seconds after jumping out of a plane.

- 14** The following formula can be used to convert Fahrenheit temperatures x to Celsius temperatures $T(x)$:

$$T(x) = \frac{5}{9}(x - 32)$$

- a** Find $T(-13)$.
- b** Find $T(86)$.
- c** Find $T^{-1}(x)$.
- d** What can the formula T^{-1} be used for?

- 15** A particle is moving along a straight line path. After t seconds, its velocity is given by the equation $v = (96t - 80)^2$.

- a** Solve for the time t at which the particle comes to rest.
- b** Determine the equation for time, t , in terms of velocity, v , that represents the motion of the particle before it has come to rest.

- 16** The tax on a new tablet is 7% of the advertised price, A .

- a** Determine the equation for the total cost T as a function of the advertised price A .
- b** Hence, express the advertised price A as a function of the total cost T .

- 17** The function $T = 2\pi\sqrt{\frac{l}{9.8}}$ can be used to find the period T of a simple pendulum of length l metres.

- a** Express the length l as a function of the period T .
- b** Hence, find the length of a pendulum whose period is 1.5 seconds. Round your answer to one decimal place.